$BN - Fe_3O_4$ Nanocomposite Based Carbon Monoxide Gas Sensor

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Course: ME398A (Undergraduate Project - I)

Introduction:

The quality of breathable air is diminishing due to the emission of various toxic gases in the earth's atmosphere. Industrialization is one of the major factors responsible for the production of these toxic gases. Carbon Monoxide is a prime pollutant gas which results from incomplete oxidation/combustion of fuels. Increasing concerns about the atmospheric CO concentration and its impact on the environment are motivating researchers to discover new materials and technologies for efficient CO sensing. Soon after the discovery of first 2D material (Graphene), it became popular for various technological applications including sensors. Its sensing application is inspired by the perfect flat structure which makes all its atoms on the surface exposed to the environment. It is reported that graphene is capable of detecting various guest molecules with atomic precision. However, its intrinsic semimetallicity restricts its path towards any practical application due to large leakage current. Nitride-based 2D materials are another category of flat materials with wider bandgap which ensures their larger acceptability for practical applications due to higher thermal and chemical stability. Owing to these properties, researchers have investigated nitride based 2D sheets and their one-dimensional counterparts for sensing applications. Boron Nitride (BN) and Aluminum Nitride (AlN) nanostructures have been predicted to exhibit potential towards sensing of CO2 and nitrogen gas. Currently, commonly used methods for the detection of CO are time-consuming and expensive. In gas-sensing applications, the sensor performance is highly dependent on the microstructure and surface area of the sensing material. Therefore, it is desirable to investigate novel materials with a high surface area for their sensing ability.

Hexagonal Boron Nitride (h-BN) has a very similar structure to that of graphene. It has a hexagonal arrangement of alternating Boron and Nitrogen atoms. Boron nitride nanosheet is a two-dimensional crystalline form of the hexagonal boron nitride (h-BN), which has a thickness of one to few atomic layers. It is similar in geometry to its all-carbon analog graphene, but has very different chemical and electronic properties – contrary to the black and highly conducting graphene, BN nanosheets are electrical insulators with a band gap of ~5.9 eV. h-BN has been found to adsorb gases such as Hydrogen, Carbon monoxide, Carbon dioxide, etc. We focused on BN because it is light and similar in structure to carbon so that high surface area BN can be synthesized. An interesting feature of h-BN is that the B-N bond has an ionic character, which may induce an extra dipole moment and facilitates adsorption of gases. Hexagonal boron nitride (h-BN) consists of heteropolar sp2-like bonding.

We analyzed the change in the electrical resistance of the proposed material with variation in CO concentration and report the sensitivity of the sensor using the measurement of Response:

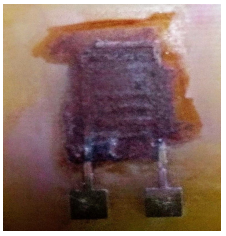
$$Response = \frac{R_g}{R_a} \times 100\,\%$$

Here, R_g is the resistance of the electrode in the presence of CO gas

R_a is the resistance of the electrode in a free atmosphere

Experimental Method:

1. Preparation of Electrode: Nanocomposite of $BN - Fe_3O_4$ was synthesized and its analysis was done. The nanocomposite was then drop-casted using ethyl alcohol to prepare an Electrode. The electrode was then cooled to room temperature.



$Fig\ 1.\ BN - Fe_3O_4\ electrode$

2. Setup and Measurement: We used Carbon Monoxide as target gas along with Nitrogen and Argon as the carrier gas. The experiments were conducted at room temperature. The flow of target gas and the carrier gas into the test chamber was controlled using a set of two mass flow controllers. Measurements for the change in resistance has been performed by using an LCR meter with a range of 200 MegaOhm resistance measurement, within the test chamber and analysis of the acquired data is performed to detect the presence and the amount of the target gas in the sample. Initially, the resistance of the sensing element in room temperature by flowing only carrier gas is recorded for creating a control experiment. This is followed by flowing the carrier gas mixed with the target gas in a series of dilution ratios and the resistance again recorded. The response was calculated using:

$$Response = \frac{R_g}{R_a} \times 100\ \%$$

$Fig\ 2.\ The\ experimental\ setup$

Results and Discussion:

1. We observed that the electrical resistance of the electrode increases as it comes in contact with the target gas.
2. The electrical resistance was found to shoot initially, then increases until a near saturation is reached.
3. The graph below (Fig 3) shows the response of the sensor with regards to varying CO concentration.

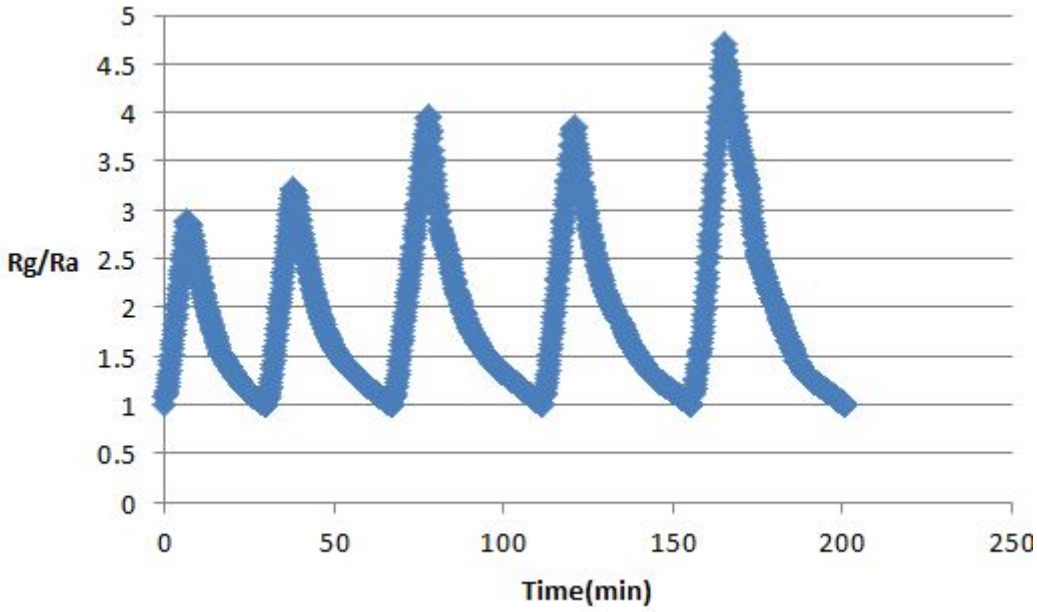


Fig 3. The response of the sensor with varying CO concentration (ppm).

4. The response time is calculated as time to reach 90% of the maximum response. Similarly, the recovery time is also calculated. The table below shows our response and recovery time observations. The recovery time is higher than the response time.

CO Concentration (PPM)	Response Time (Sec)	Recovery Time (Sec)
10	350	990
20	410	1260
50	560	1410
70	490	1430
100	530	1380

Table 1. Recovery and Response time for varying CO Concentration.

5. The response and recovery time of the proposed sensor is high compared to some of the already existing sensing materials as mentioned in Table 2.

Material	Concentration of CO (PPM)	Response Time (Sec)	Recovery Time (Sec)
$rGO-\alpha-Fe_2O_3$	10	21	8
$Au-\alpha-Fe_2O_3$	1000	140	140
$\alpha-Fe_2O_3$	100	2-8	91-173
rGO	10	30	900
$BN-Fe_3O_4$	10	350	990

Table 2. Response and Recovery time of Some CO Sensors.

Conclusion:

In summary, we have successfully synthesized $BN-Fe_3O_4$ nanocomposites and studied their response behavior to changes in the concentration of CO. CO sensors are very useful for monitoring of a variety of gases within mines, automotive exhausts, chemical plants, etc. The response of the proposed material does not show promising CO sensing characteristics. For more files/data related to experiment click here.

Acknowledgment:

I would like to thank my project supervisor Dr. Niraj Sinha and the Biomedical Research Lab, IIT Kanpur for providing me with this great opportunity. This has been an exceptional learning experience wherein I got a chance to explore some basic concepts in sensing technology and nanomaterials. I would also like to thank Mr. Raghubeer Bangari (Ph.D. Student) for helping me whenever I was stuck. This project would not have been possible without his mentorship.

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